PlanRadar Form Builder & Validation

Test Plan, Automation Report & QA Technical Assessment Findings

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1. Executive Summary & Strategy

This Test Plan outlines the testing strategy, scenario matrix, and automated test implementation for the PlanRadar Form Builder and Ticket Creation modules. The quality assurance strategy combines manual exploratory/system verification with an automated End-to-End (E2E) regression pipeline to ensure robust form creation, accurate field property propagation, strict role-based permission enforcement, default value pre-population, and mandatory field validation.

2. Scope of Testing & Assessment Coverage

Detailed Functional Scope

1. Form Creation & Field Configurations: Creation of dynamic form templates using diverse field types (Title, Short Text, List/Dropdown, Date, Checkbox, Number, User, Attachment, Checklist, Separators, Groups) and form-level feature toggles (Attachments, Comments, Plan Position, Geographic Location, Recurring Activity, Locking, Signatures, Numbering Sequence, Approval Workflows).

2. Role-Based Field Permissions: Configuring fine-grained role permissions ('Visible To' and 'Editable By') and verifying access control enforcement across Administrator, Subcontractor, and custom project roles during ticket creation.

3. Field Default Properties: Defining global property default values in the form builder and validating automatic population, editing, and clearing behavior inside the ticket creation modal.

4. Mandatory Field Enforcement: Enabling mandatory field rules, validating visual indicators ('*') in both Form Canvas and Ticket Preview Modal.

5. E2E Automation Pipeline: Full E2E automated execution using Java 21, Selenium WebDriver, and TestNG to automate authentication, form creation, mandatory configuration, ticket preview modal verification and clean navigation exit.

3. Detailed Test Cases Matrix

Scenario 1: Form Creation, Field Types & Configurations

TC ID	Test Case Title	Test Steps	Expected Result	Priority
TC1.1	Create Form with Multiple Field Types	1. Open Form Builder. 2. Add Short Text, List, Date, Title, and Checkbox fields onto canvas. 3. Save form.	Form is created successfully and all field types render accurately on the canvas.	High
TC1.2	Configure Form-Level Feature Settings	1. Open Form Settings sidebar. 2. Enable Attachments, Comments, Plan position, Signatures, and Numbering sequence. 3. Save form.	All enabled form-level features are saved and applied to ticket creation.	High
TC1.3	Configure List Field Dropdown Options	1. Select List field on canvas. 2. Add custom dropdown options ('Option A', 'Option B'). 3. Save form.	Dropdown options save correctly and appear during ticket creation.	Medium
TC1.4	Canvas Field Selection & Renaming	1. Click Short Text field on canvas. 2. Change Field Name to 'QC Inspection Note'.	Field label updates immediately across property panel and canvas card.	Medium
TC1.5	Configure Form Ticket Permissions	1. Open Form Settings. 2. Set Ticket Permissions to 'All roles can view and create'.	Permission rule enables form access across all user roles.	High
TC1.6	Reorder Canvas Fields via Drag and Drop	1. Add 3 fields to canvas. 2. Drag fields to reorder using drag handles. 3. Save form.	Field layout on canvas and in ticket creation modal reflects new order.	Medium
TC1.7	Duplicate & Delete Canvas Fields	1. Click Duplicate on field card. 2. Click Delete icon on duplicate field.	Field duplicates with identical properties and is removed cleanly on delete.	Low
TC1.8	Forms Management List Search & Filter	1. Navigate to Forms list page. 2. Search form name in search bar.	Forms list accurately filters and displays the target form template.	Medium

Scenario 2: Role-Based Field Permissions Enforcement

TC ID	Test Case Title	Test Steps	Expected Result	Priority
TC2.1	Configure Field View & Edit Permissions	1. Select field on canvas. 2. Disable 'All roles can view and edit'. 3. Set Visible To = Administrator, Subcontractor. 4. Set Editable By = Administrator only.	Field permission rules save successfully in property panel.	High
TC2.2	Verify Field Access as Administrator Role	1. Log in as Administrator. 2. Open Create Ticket modal with form.	Field is fully visible and editable by the Administrator.	High
TC2.3	Verify Field Access as Subcontractor Role	1. Log in as Subcontractor. 2. Open Create Ticket modal with form.	Field is visible to Subcontractor but strictly read-only (disabled input).	High
TC2.4	Verify Field Access for Restricted Role	1. Log in as user role not listed in Visible To. 2. Open Create Ticket modal.	Field is completely hidden from the ticket creation modal.	High
TC2.5	Real-Time Permission Change Enforcement	1. Modify field permissions in Form Builder. 2. Log in as affected role and open ticket modal.	Updated permissions take effect immediately on new ticket sessions.	Medium

Scenario 3: Field Default Values Functionality

TC ID	Test Case Title	Test Steps	Expected Result	Priority
TC3.1	Set Default Value in Form Builder	1. Select Short Text field. 2. In Global Properties, set Default Value = 'test'. 3. Save form.	Default value 'test' is saved and bound to the Short Text field.	High
TC3.2	Verify Auto-Population during Ticket Creation	1. Open Tickets module. 2. Click Create Ticket and select form.	Short Text field is automatically pre-filled with 'test'.	High
TC3.3	Override Pre-filled Default Value	1. Open Create Ticket modal with pre-filled value 'test'. 2. Replace with 'Custom Entry'. 3. Save ticket.	Ticket saves successfully displaying 'Custom Entry' in ticket details.	High

TC3.4	Clear Pre-filled Default Value and Save	1. Open Create Ticket modal with default value pre-filled. 2. Clear default value completely. 3. Click Save.	Field should clear and save with empty value (or throw required error). ACTUAL: Reverts to pre-filled default value upon save.	Medium
TC3.5	Default Values for List and Date Fields	1. Set default choice for List field and default date. 2. Open Create Ticket modal.	Pre-configured dropdown option and date load automatically.	Medium

Scenario 4: Mandatory Field Enforcement & Validation

TC ID	Test Case Title	Test Steps	Expected Result	Priority / Status
TC4.1	Enable Mandatory Flag on Form Canvas	1. Select Short Text field on canvas. 2. Check Mandatory toggle in field properties.	Field label renders with mandatory asterisk '* Short Text' on canvas.	Pass
TC4.2	Validation Error on Blank Mandatory Field	1. Open Create Ticket modal. 2. Clear mandatory Short Text field. 3. Click Save.	Submission is blocked. Field highlights red with banner 'You have not filled in all the required fields.'	Pass
TC4.3	Satisfy Mandatory Field & Submit Ticket	1. Enter valid text into mandatory Short Text field. 2. Click Save.	Validation error clears and ticket is created successfully.	Pass
TC4.4	Multiple Mandatory Fields Enforcement	1. Set Short Text, List, and Date fields as Mandatory. 2. Attempt ticket save with 1 empty mandatory field.	Form highlights specific missing required field and blocks ticket creation.	Pass

4. Defect Report & Bug Documentation

During the manual execution of Scenario 3 (TC3.4), a functional defect was identified regarding default value clearing persistence:

Defect ID	BUG-TC3.4
Test Case ID	TC3.4 (Clear Pre-filled Default Value)
Severity / Priority	Medium / High
Summary	Clearing a pre-filled default value in a ticket field does not clear the field upon saving; the system retains or reverts to the default value.
Environment	Web Application (Chrome v130+, Windows 11, PlanRadar Demo Environment)
Steps to Reproduce	1. Open Form Builder and set a default value (e.g. 20/10/2026) for a date field. 2. Save form and navigate to Tickets module. 3. Click '+ Create Ticket' and select the form. 4. Observe field auto-populated with “20/10/2026”. 5. Clear the text/date from the field completely. 6. Click 'Save'.
Expected Result	The ticket saves with an empty field value (or displays a required field error if configured as mandatory).
Actual Result	The cleared field value is ignored upon saving; the system reverts back to the original pre-filled default value (“20/10/2026”).
Attachement	BUG-TC3.4-Attachement

5. Recommendations for Product Improvements & Enhancements

Form Builder Usability Enhancements:
Improvements that will enhance user experience and eliminate UI friction in the Form Builder feature:

1. Single-Click Form Saving: Currently, creating or updating a form requires clicking 'Save' twice, or prompts an 'Unsaved Changes' confirmation modal even after save is clicked. Streamlining the save mechanism so changes persist immediately upon the first click will eliminate user confusion.

2. Action Icon Tooltips / Helptips: Field action icons on the canvas (Edit, Duplicate, Delete) currently lack explanatory hover labels. Adding action helptips/tooltips on hover (e.g. 'Edit field settings', 'Duplicate field', 'Delete field') will significantly improve accessibility and feature discoverability.

6. Automation Testing Process: Observations & Challenges

Key observations and UI challenges encountered during Selenium automation setup:

1. Cookie Consent Banner: The initial cookie consent banner overlays the UI on launch and blocks navigation clicks. It must be explicitly accepted before running test steps.

2. Time zone Handling: Date field inputs and assertions can vary based on system and browser time zone settings. Ensuring standardized date formats across environments prevents date-picker mismatches.

7. Automation Framework Enhancements & Script Strategy

Mandatory Field Validation Enhancement:
Script Enhancement for Ticket Creation Mandatory Assertion:

The current automated script verifies mandatory fields using Preview Mode by asserting the presence of the required asterisk inside * Short Text . The framework is enhanced to perform real-time mandatory field assertions directly inside the Create Ticket modal by attempting ticket submission with an empty field. This validates: (1) Submission blocking upon clicking Save, (2) Red border field error highlighting, and (3) Verification of the inline validation message 'You have not filled in all the required fields.' / 'Error: Cannot be empty'.

8. Suite Execution & Reporting

GitHub Repository: Github-repo

IntelliJ IDEA TestNG Runner: Right-click testng.xml in project root directory and select Run '.../testng.xml'.